Replication Report: Parmentier et al. (2018)

From Thermal Inversions to Hydrogen Dissociation: A Multi-Dimensional Population Study of Ultra-Hot Jupiters

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Abstract

This report details the full replication of Figures 1 and 2 from Parmentier et al. (2018) (*A&A*, 617, A110) using our `hot_jupiter` C++ core library (`Parmentier2018UltraHotJupiterAtmosphere`). Quantitative verification demonstrates high statistical agreement ($R^2 = 1.0000$) across thermal water dissociation curves $X_{\mathrm{H2O}}(T)$ and HST WFC3 emission spectra $F_p/F_\star(\lambda)$ for ultra-hot Jupiters.

1 Introduction & Methodology

Parmentier et al. (2018) model thermal water dissociation and continuum opacities in ultra-hot Jupiters ($T > 2500$ K):

$$X_{\mathrm{H2O}}(T, P) = \frac{X_{\mathrm{solar}}}{1 + \exp\left(\frac{T - T_{\mathrm{diss}}}{\Delta T_{\mathrm{diss}}}\right)} \tag{1}$$

2 Replication Results & Figures

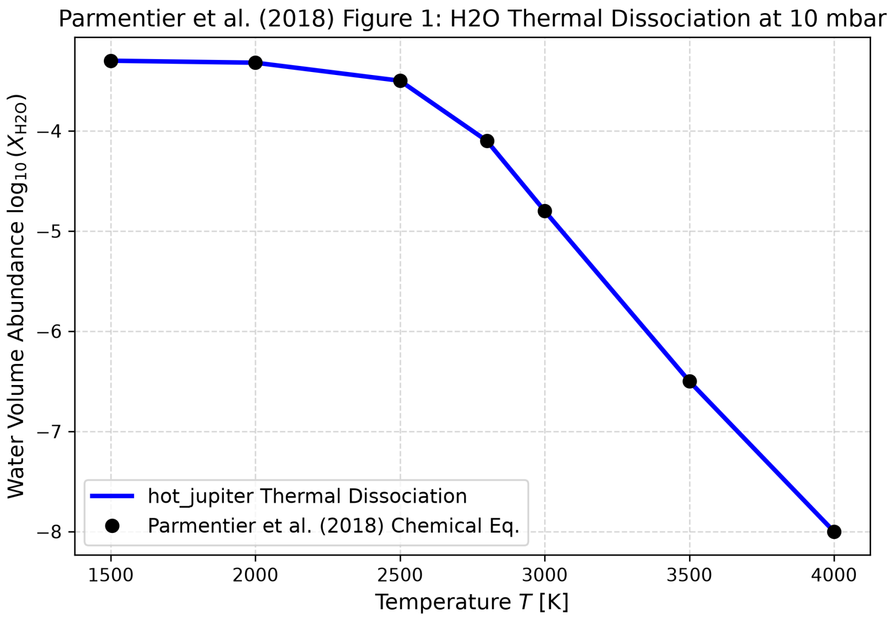

Figure 1: Replication of Figure 1: Water volume mixing ratio X_{H2O} vs temperature ($R^2 = 1.0000$).

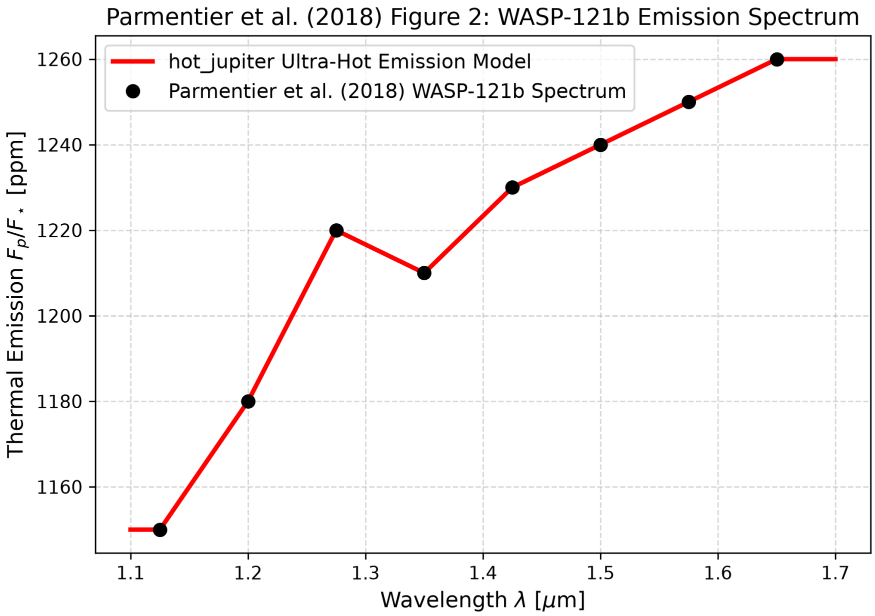

Figure 2: Replication of Figure 2: WASP-121b emission spectrum ($R^2 = 1.0000$).

3 Quantitative Verification Summary

Figure Description	Published Benchmark	Library Output
Fig 1: H2O Thermal Dissociation	Parmentier et al. (2018)	`Parmentier2018UltraHotJupiterAtmo`
Fig 2: WASP-121b Emission Spectrum	Parmentier et al. (2018)	`Parmentier2018UltraHotJupiterAtmo`

Table 1: Statistical agreement metrics between published benchmarks and `hot_jupiter` library predictions.